

Translated

From: e-maxx.ru

Fibonacci Numbers

The Fibonacci sequence is defined as follows:

$$F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$$

The first elements of the sequence ([OEIS A000045](#)) are:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

Properties

Fibonacci numbers possess a lot of interesting properties. Here are a few of them:

- Cassini's identity:

$$F_{n-1} F_{n+1} - F_n^2 = (-1)^n$$

This can be proved by induction. A one-line proof by Knuth comes from taking the determinant of the 2x2 matrix form below.

- The "addition" rule:

$$F_{n+k} = F_k F_{n+1} + F_{k-1} F_n$$

- Applying the previous identity to the case $k = n$, we get:

$$F_{2n} = F_n (F_{n+1} + F_{n-1})$$

- From this we can prove by induction that for any positive integer k , F_{nk} is multiple of F_n .
- The inverse is also true: if F_m is multiple of F_n , then m is multiple of n .
- GCD identity:

$$\text{GCD}(F_m, F_n) = F_{\text{GCD}(m, n)}$$

- Fibonacci numbers are the worst possible inputs for Euclidean algorithm (see [Lame's theorem](#) in [Euclidean algorithm](#))